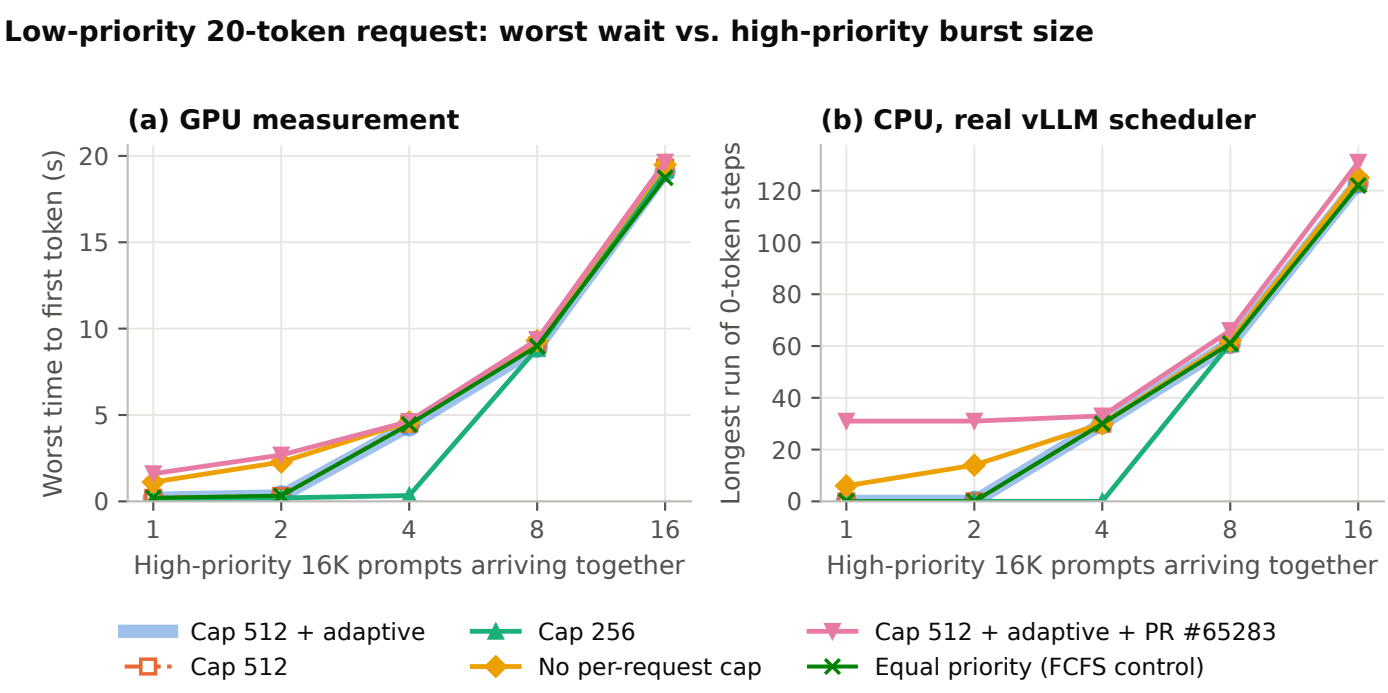
Low-priority 20-token request: worst wait vs. high-priority burst size
(a) GPU measurement
Worst time to first token (s)
0
5
10
15
20
1
2
4
8
16
High-priority 16K prompts arriving together
(b) CPU, real vLLM scheduler
Longest run of 0-token steps
0
20
40
60
80
100
120
1
2
4
8
16
High-priority 16K prompts arriving together
Cap 512 + adaptive
Cap 512
Cap 256
No per-request cap
Cap 512 + adaptive + PR #65283
Equal priority (FCFS control)